Run ID: berlin52_1791575927_76bfc3

Adaptive Probe-Guided Genetic Algorithm (APGGS) — Experiment Report

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Engine Version: v4.2

1. Executive Summary & Dataset Metadata

Dataset Name	BERLIN52	**Cities / Nodes**	52
Edge Weight Type	EUC_2D	**Known Best Optimum**	7,542
Final Global Best Cost	7,542.00	**Optimality Gap**	+0.0000%
Best Found Generation	15	**Completed Generations**	15
Stagnation Streak	0 / 30	**Stopping Condition**	30 gens without improvement
Experiment Status	COMPLETED	**Total Execution Time**	1.97 s
Stop Reason	0.0% gap achieved (Known optimum found)	**Random Seed**	1

2. Initial High-Quality Route Construction

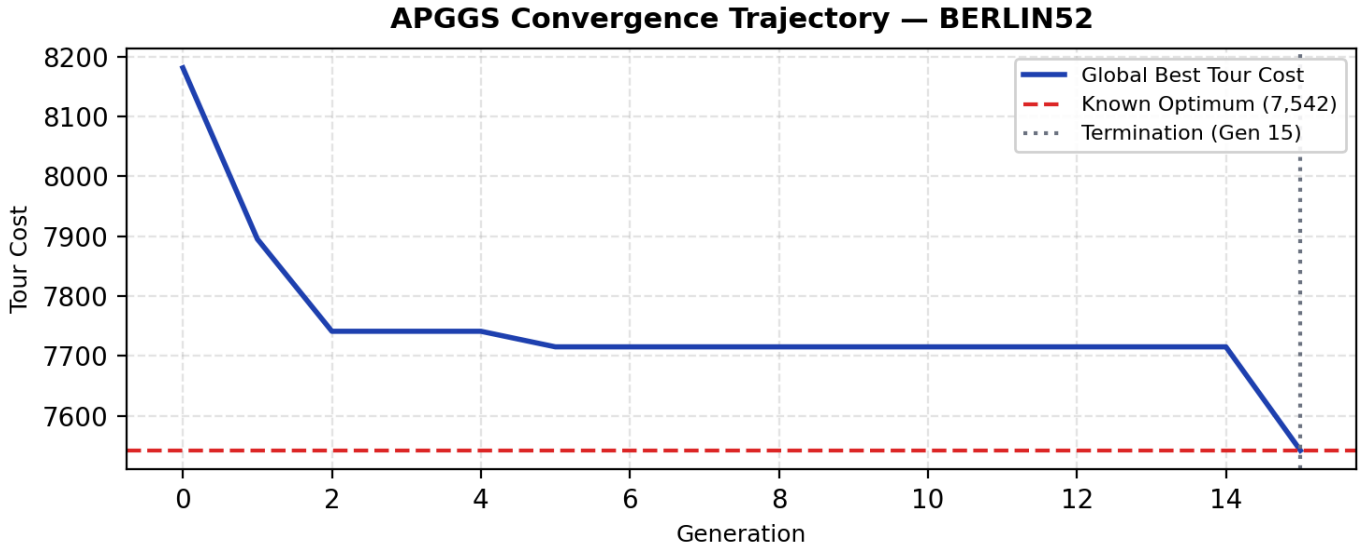
Construction Stage	Resulting Tour Cost	Improvement vs Prev	Duration (s)
Stage 1: Greedy Nearest-Neighbor Tour	9,951.00	Baseline	0.0011
Stage 2: 2-Opt Local Search Improvement	9,688.00	-263.00	0.0001
Stage 3: 2-Opt Deep Refinement Seed	8,995.00	-693.00	0.0002
Stage 4: Initial 70-Individual Population (NN+Greedy+Random)	8,181.00	Population Seeded	0.0350

3. Genetic Algorithm & Adaptive Pipeline Configuration

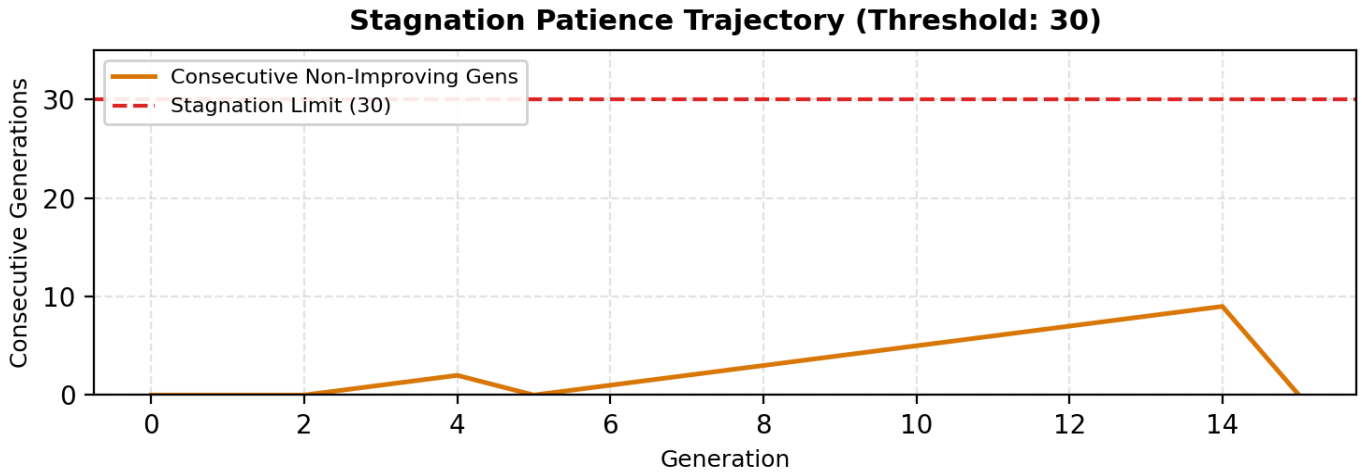
Parameter	Configured Value	Parameter	Configured Value
Population Size	70	Mutation Rate	0.20
Nearest-Neighbor Seeds	17	Swap Fraction	0.02
Greedy Edge Seeds	11	Segment Crossover Prob	0.40
Random Tour Seeds	42	Offspring 2-Opt Prob	0.35
K-Nearest Neighbors	15	Adaptive Elites Count	14
Tournament Size	4	PMG Memory Decay	0.92
Selective LKH Cadence	Every 15 gens	Stagnation Window	15
Selective LKH Individuals	3	Stagnation Patience Limit	30 consecutive gens

4. Research Visualizations & Trajectory Curves

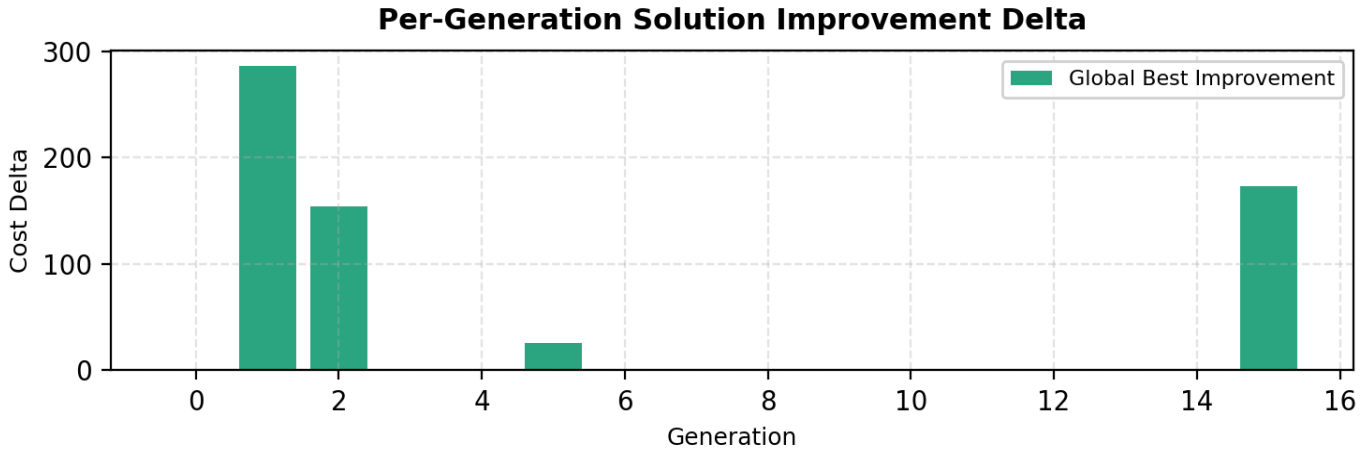
Convergence Trajectory: Global Best Cost vs Generation

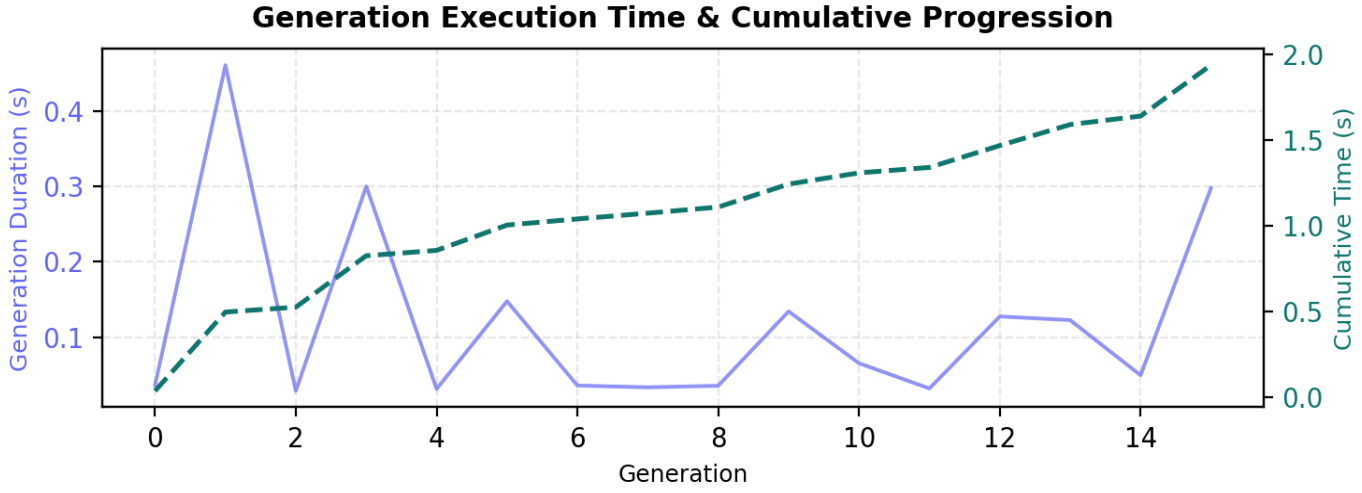


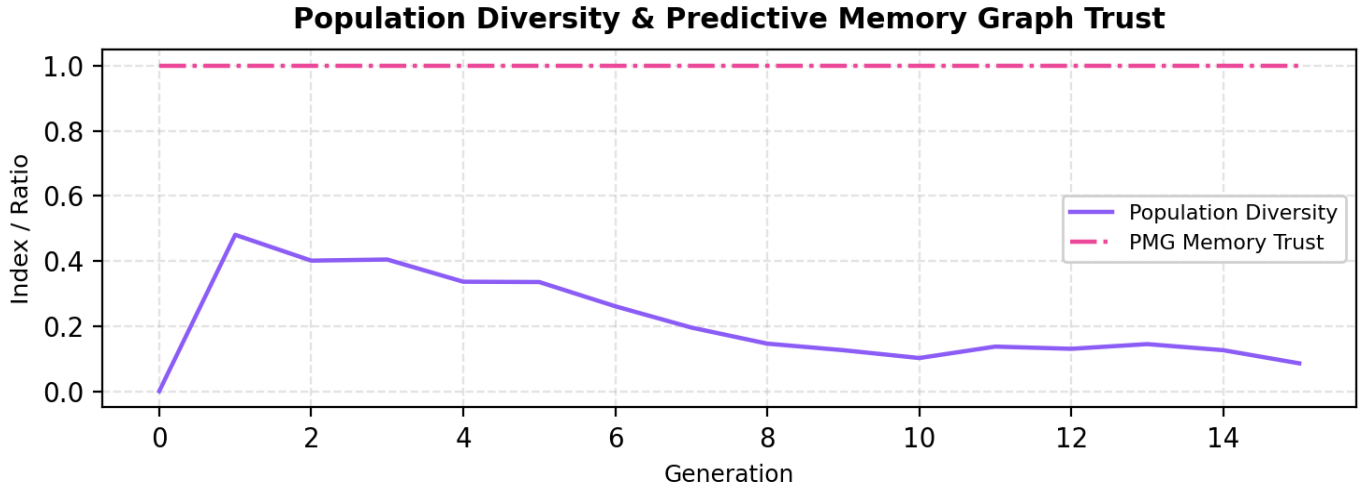
Stagnation Counter Progression (Target: 30 Consecutive Non-Improving Gens)



Per-Generation Cost Improvement & Runtime Progression







5. Evolution Highlights (Global-Best Discoveries)

Gen	Global Best Cost	Cost Reduction	Gap (%)	Operator / Method	PMG Trust	Non-Imp Streak	Duration (s)
0	8,181.00	0.00	+8.4726%	Nearest Neighbor (start=39)	1.00	0	0.0363
1	7,895.00	286.00	+4.6805%	Elite + predictive_or_opt	1.00	0	0.4606
2	7,741.00	154.00	+2.6386%	OX Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	0.0287
5	7,715.00	26.00	+2.2938%	Segment Crossover + predictive_or_opt	1.00	0	0.1477
15	7,542.00	173.00	+0.0000%	Elite + Selective LKH	1.00	0	0.2976

6. Final Discovered Tour & Solution Vector

Optimized Tour Length: 7,542.00 | **Optimality Gap:** +0.0000% | **Discovered at Generation:** 15

Permutation Sequence Preview:

```
0 → 48 → 31 → 44 → 18 → 40 → 7 → 8 → 9 → 42 → 32 → 50 → 10 → 51 → 13 → 12 → 46 → 25 → 26 → 27 → 11 → 24 → 3 → 5 → 14 → 4 → 23 → 47 → 37 → 36 → 39 → 38 → 35 → 34 → 33 → 43 → 45 → 15 → 28 → 49 → ... [12 additional cities] → 0
```

7. Execution Event Log & Diagnostics

- Initialized berlin52 (52 cities). Initial best cost: 8,181.00
- Gen 1: Discovered new global-best 7,895.00 (-286.00)
- Gen 2: Discovered new global-best 7,741.00 (-154.00)
- Gen 5: Discovered new global-best 7,715.00 (-26.00)
- Gen 15: Discovered new global-best 7,542.00 (-173.00)
- Gen 15: 0.0% optimality gap achieved. Experiment completed.

8. Artifact Manifest & Research Reproducibility

All raw experimental records have been saved to runs/berlin52_1791575927_76bfc3/ including summary.json, generation_history.jsonl, best_route.csv, and report.pdf.